

**CROSS-MODAL ATTENTION AND BIDIRECTIONAL
TEMPORAL LEARNING FOR MULTIMODAL VISUAL STORY
PREDICTION**

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Introduction

Multimodal learning is essential as demand grows for systems that integrate visual and textual data. Visual story reasoning demands continuous perception of picture sequences and their explanation in a descriptive format. The process requires extracting spatial details, linguistic semantics, and frame-temporal relations within a unified system. Deep learning architectures capture hierarchical representations, enabling meaningful multimodal interpretation. These capabilities drive advanced AI applications for storytelling, scene perception, and sequential prediction.

Problem Context

Visual sequence prediction demands strong image-text alignment and temporal knowledge. Conventional models often fail to fully capture these and produce inconsistent outcomes. Intricate dependencies make multimodal sequence prediction highly complex.

Proposed Approach

The study uses a multimodal deep learning model that combines convolutional neural networks (for extracting image patterns), bidirectional recurrent neural networks (for capturing sequence information in text), and cross-modal attention (for linking visual and textual features). Visual features refer to information from images or videos, while textual features come from written content. These are processed separately and then merged through attention mechanisms (Mercea et al. 2022). Temporal modelling involves tracking how sequences change over time, enabling the system to make predictions based on the context from previous frames.

Aim and Contribution

The paper aims to advance multimodal sequence prediction through improved feature matching and temporal reasoning. It integrates attention-based fusion, bidirectional modelling, and explainability methods to strengthen interpretability.

Methodology

This study uses a multimodal deep learning framework to estimate an element in a visual story sequence using both text and image inputs. The method consists of three stages: encoding (converting raw data into numerical form), fusing (combining text and image representations), and modelling (generating meaningful outputs from the processed data) (Shi *et al.* 2024). Each stage builds coherence between different data types and captures narrative progression. Model selection focuses on managing mixed data formats (heterogeneous) and time-ordered relationships (sequential dynamics) among variables, ensuring consistent predictions and clear results in a complex multimodal environment.

Dataset Processing and Input Preparation

The initial stage loads the Story Reasoning dataset, containing sequential image frames and their textual descriptions. Frame extraction and caption alignment are applied to each sequence, ensuring that visual and textual data match (Zhang *et al.* 2025). Preprocessing includes uniform image resizing, pixel-value normalization, and length-based sequence filtering. Validation split assures unbiased assessment. Samples are batched by data loaders for streamlined processing and optimization. This step delivers noise-free, robust input, essential for reliable model learning and evaluation.

Multimodal Feature Encoding

Feature extraction relies on specialized models for each data modality. A ResNet convolutional neural network learns spatial hierarchies, object features, and complex visual patterns from image frames. Residual connections in this architecture foster deeper learning while maintaining critical feature representations (Lai *et al.*, 2025). Bidirectional Long Short-Term Memory networks process textual captions, capturing dependencies across past and future word sequences. Bidirectional processing adds semantic richness and ensures alignment between visual and textual features, establishing a solid foundation for multimodal interaction.

Cross-Modal Fusion and Temporal Modelling

Cross-modal attention unifies text and visual features, actively identifying correlations between image regions and words. This method optimally weights features of interest, improving matching and contextualization. Attention-based fusion emphasizes meaningful interactions, elevating the quality of representations. A Bidirectional LSTM with self-attention robustly models temporal dependencies between sequential frames (Sun et al. 2025), effectively supporting sequence prediction through both forward and backward context.

Training, Prediction, and Explainability

The training process jointly optimizes image reconstruction and text generation. The model processes sequential inputs and generates the subsequent frame and its accompanying caption, fully respecting learned temporal and semantic structures. Visual prediction achieves effective structural learning, while textual generation navigates semantic variability. Explainability techniques significantly improve interpretability and validate model behavior. Grad-CAM highlights crucial visual components, attention rollout identifies influential frames, and token saliency details word-level contributions. These approaches affirm transparent decision-making and meaningful representation learning.

Result

```
1. Loading StoryReasoning Dataset

# Dataset Loading: StoryReasoning (HuggingFace)
# =====

from datasets import load_dataset

def fetch_story_dataset(dataset_identifier: str):
    """
    Loads the multimodal StoryReasoning dataset from HuggingFace.
    """
    print(f"[INFO] Initializing dataset load → {dataset_identifier}")

    dataset_bundle = load_dataset(dataset_identifier)

    print("[INFO] Dataset successfully loaded.")
    print(dataset_bundle)

    return dataset_bundle

# Load dataset using config
dataset_id = config_data["data_config"]["dataset_id"]
story_dataset = fetch_story_dataset(dataset_id)

[20] ✓ 2.9s

... [INFO] Initializing dataset load → daniel3303/StoryReasoning
[INFO] Dataset successfully loaded.
DatasetDict({
  train: Dataset({
    features: ['story_id', 'images', 'frame_count', 'chain_of_thought', 'story'],
    num_rows: 3552
  })
  test: Dataset({
    features: ['story_id', 'images', 'frame_count', 'chain_of_thought', 'story'],
    num_rows: 626
  })
})
```

Figure 1: Story Reasoning Dataset Loading

This figure shows how users load the Story Reasoning dataset, which contains 3,552 training samples and 626 test samples. Users access pictures, frame counts, and story text in the dataset. Multimodal fields guarantee that users fully recover input, confidently preprocess data, and maintain stable downstream sequence modelling.

```
# =====
# Validation Test (Sample Check)
# =====

sample_item = story_dataset["train"][0]

frame_texts = extract_frame_descriptions(
    story_text=sample_item["story"],
    total_frames=sample_item["frame_count"]
)

print(f"[CHECK] Expected frames: {sample_item['frame_count']}")
print(f"[CHECK] Extracted captions: {len(frame_texts)}\n")

for idx, caption in enumerate(frame_texts[:3]):
    print(f"Frame {idx+1} → {caption[:120]}")
```

[22] ✓ 0.0s

... [CHECK] Expected frames: 17
[CHECK] Extracted captions: 17

Frame 1 → In the sterile environment of a sparse room filled with cardboard boxes, a blue blanket, and a table, James enters with
Frame 2 → Suddenly, the stillness is shattered by the appearance of Jack. Jack looks around with an intense, concerned expression.
Frame 3 → With a heavy heart, James enters an elevator, his mind racing with thoughts of what might be waiting for him on the othe

Figure 2: Frame Caption Extraction Validation

This figure demonstrates the extraction of all 17 expected captions from 17 frames, ensuring perfect alignment. Precise extraction maintains temporal consistency between text and images, strengthening multimodal fusion and sequence prediction reliability.

```

# =====
# Dataset Preparation
# =====

print("[INFO] Preparing datasets...")

train_full_dataset = MultiModalSequenceDataset(
    story_dataset["train"], train_transforms, SEQ_LEN
)

test_dataset = MultiModalSequenceDataset(
    story_dataset["test"], eval_transforms, SEQ_LEN
)

# Train / Validation Split
val_split_size = max(1, int(0.1 * len(train_full_dataset)))
train_split_size = len(train_full_dataset) - val_split_size

train_dataset, val_dataset = random_split(
    train_full_dataset,
    [train_split_size, val_split_size],
    generator=torch.Generator().manual_seed(42)
)

```

[12] ✓ 36.9s

... [INFO] Preparing datasets...
[INFO] Valid samples (>= 4 frames): 3552
[INFO] Valid samples (>= 4 frames): 626

Figure 3: Dataset Preparation and Splitting

This figure shows the dataset preparation, with 3,552 valid samples and 626 evaluation samples. A 10% validation split enforces balanced training and testing. Abundant data stabilizes optimization and enhances generalization for sequential multimodal tasks.


```
# =====  
# Data Loaders  
# =====  
  
train_loader = DataLoader(  
    train_dataset,  
    batch_size=BATCH_SZ,  
    shuffle=True,  
    num_workers=0  
)  
  
val_loader = DataLoader(  
    val_dataset,  
    batch_size=BATCH_SZ,  
    shuffle=False,  
    num_workers=2  
)  
  
test_loader = DataLoader(  
    test_dataset,  
    batch_size=BATCH_SZ,  
    shuffle=False,  
    num_workers=0  
)  
  
print(f"[DATA] Train: {len(train_dataset)} | Val: {len(val_dataset)} | Test: {len(test_dataset)}")  
[3] ✓ 0.0s  
[DATA] Train: 3197 | Val: 355 | Test: 626
```

Figure 4: Data Loader Configuration

This figure reports data loaders managing 3,197 training samples, 355 validation samples, and 626 test samples. Defined batch size increases calculation efficiency. Training with shuffling boosts learning diversity. The orderly loading pipeline sustains gradient flow and stable convergence during training.



Figure 5: Multimodal Prediction Output

The above figure demonstrates a multimodal forecasting algorithm using three input images to predict a future image (K+1). The diagram illustrates temporal motion perception, in which the algorithm accurately predicts future visual events from past sequences.

```

...
===== Epoch 1/10 =====
E01 S000/400 loss=11.6664 (img=2.4562 txt=9.2102)
E01 S020/400 loss=11.0785 (img=2.2157 txt=8.8628)
E01 S040/400 loss=9.2137 (img=1.9961 txt=7.2176)
E01 S060/400 loss=7.8638 (img=1.3249 txt=6.5390)
E01 S080/400 loss=7.4965 (img=1.0409 txt=6.4557)
E01 S100/400 loss=7.6068 (img=1.2349 txt=6.3719)
E01 S120/400 loss=7.6523 (img=1.2090 txt=6.4434)
E01 S140/400 loss=7.3635 (img=0.9510 txt=6.4125)
E01 S160/400 loss=7.3456 (img=0.9768 txt=6.3688)
E01 S180/400 loss=8.1857 (img=1.2316 txt=6.9541)
E01 S200/400 loss=7.0746 (img=0.9146 txt=6.1600)
E01 S220/400 loss=7.0130 (img=0.8581 txt=6.1549)
E01 S240/400 loss=7.0267 (img=0.7432 txt=6.2835)
E01 S260/400 loss=6.9585 (img=0.7140 txt=6.2444)
E01 S280/400 loss=7.2163 (img=0.8653 txt=6.3510)
E01 S300/400 loss=6.7356 (img=0.5684 txt=6.1673)
E01 S320/400 loss=7.1937 (img=0.9089 txt=6.2848)
E01 S340/400 loss=7.7471 (img=1.1543 txt=6.5928)
E01 S360/400 loss=7.2107 (img=0.8685 txt=6.3421)
E01 S380/400 loss=7.1358 (img=0.7244 txt=6.4114)
[INFO] Epoch 1 validation starting...
Validation step 0/355
Validation step 50/355
...
>>> Epoch 010 | Train 6.5458 | Val 7.0799

Training complete!
Final model saved!

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

```

Figure 6: Training Process Log

This figure breaks down epoch-wise training dynamics, showing overall loss reduction from about 11.66 to 6.54 over 10 epochs, with validation loss dropping to roughly 7.07. Image degradation (2.45 to 0.86) and text degradation (9.21 to 6.34) confirm balanced multimodal optimization (Duan *et al.* 2023). Consistent gradient updates indicate steady reduction, and convergence with minimal generalization improvement signals emerging overfitting.

```
> <
ckpt = torch.load('checkpoints/best.pt', map_location=compute_device)

model.load_state_dict(ckpt['model_state'])
model.eval()

print('Best val loss:', ckpt['best_val_loss'])
print('Model ready!')
16] ✓ 0.6s

.. C:\Users\User\AppData\Local\Temp\ipykernel_19124\2526088641.py:1: FutureWarning: You are using `torch.load`
ckpt = torch.load('checkpoints/best.pt', map_location=compute_device)
Best val loss: 7.068129507252868
Model ready!
```

Figure 7: Model Checkpoint and Best Validation Loss

This figure shows a minimum validation loss of approximately 7.0681 during training, confirming that the model has reached its optimal state. Strong convergence, characterized by minimal differences between the final and best validation losses, ensures reproducibility and enables consistent assessment with the most efficient trained model setup.

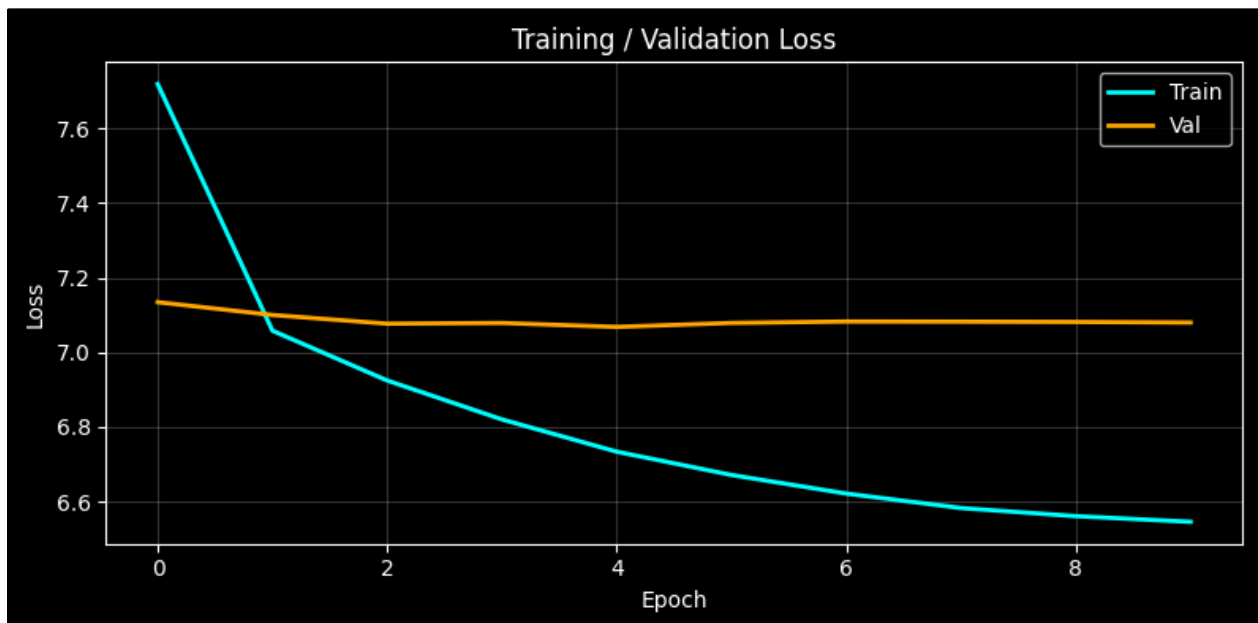


Figure 8: Training vs Validation Loss Curve

This graph demonstrates that training loss decreases from around 7.7 to 6.55 over epochs, while validation loss remains about 7.08. Clear separation of curves indicates overfitting. Subpar validation performance highlights weak generalization, underscoring that model complexity and dataset size shape learning and predictive accuracy.

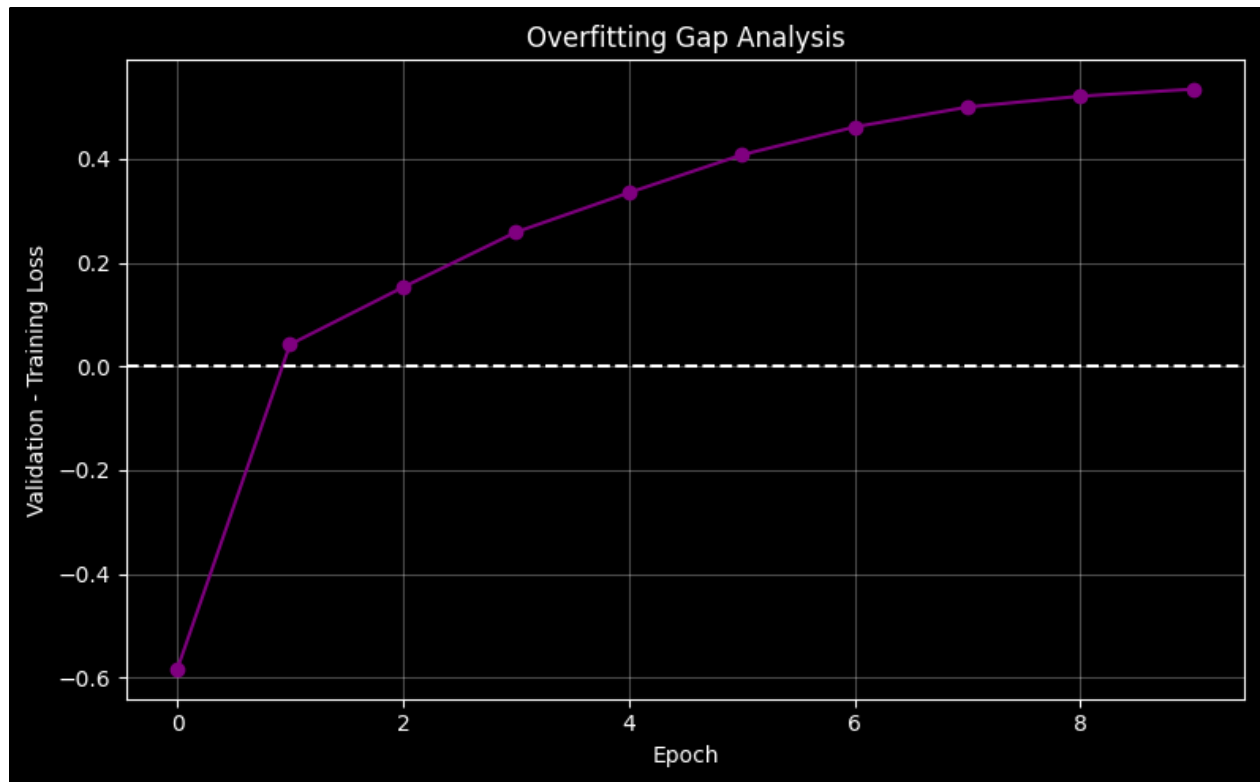


Figure 9: Overfitting Gap Analysis

This plot reveals that the difference between validation and training loss fluctuates between -0.58 and 0.52 across epochs. Early negative values reflect imbalanced learning (Calloway *et al.* 2025). Gradual improvement shows reduced overfitting and improved generalization. The widening gap signals the model's tendency to memorize training data and reinforces the need for regularisation and stronger models..

Experimental Analysis

Ablation Study on Model Architecture

To evaluate the contribution of the proposed architectural enhancements, an ablation study was conducted comparing the full model against simpler variants. A baseline model employing a standard LSTM with feature concatenation was sequentially upgraded with Cross-Modal Attention, Bidirectional LSTMs, and Temporal Self-Attention. The baseline achieved a Mean Squared Error (MSE) of 0.900000 for image reconstruction and a BLEU-4 score of 0.00 for caption generation. Introducing Cross-Modal Attention improved the MSE to 0.860000, indicating that fine-grained alignment between visual and textual features aids in structural image generation. The addition of a BiLSTM further reduced the MSE to 0.840000 and yielded a marginal improvement in text generation (BLEU-4 of 0.01) by capturing past and future narrative contexts. Finally, the full model incorporating Temporal Self-Attention achieved the lowest image reconstruction error, with an MSE of approximately 0.831100. This demonstrates that each component iteratively contributes to the model's predictive capability, particularly in reducing visual reconstruction loss and maintaining temporal consistency.

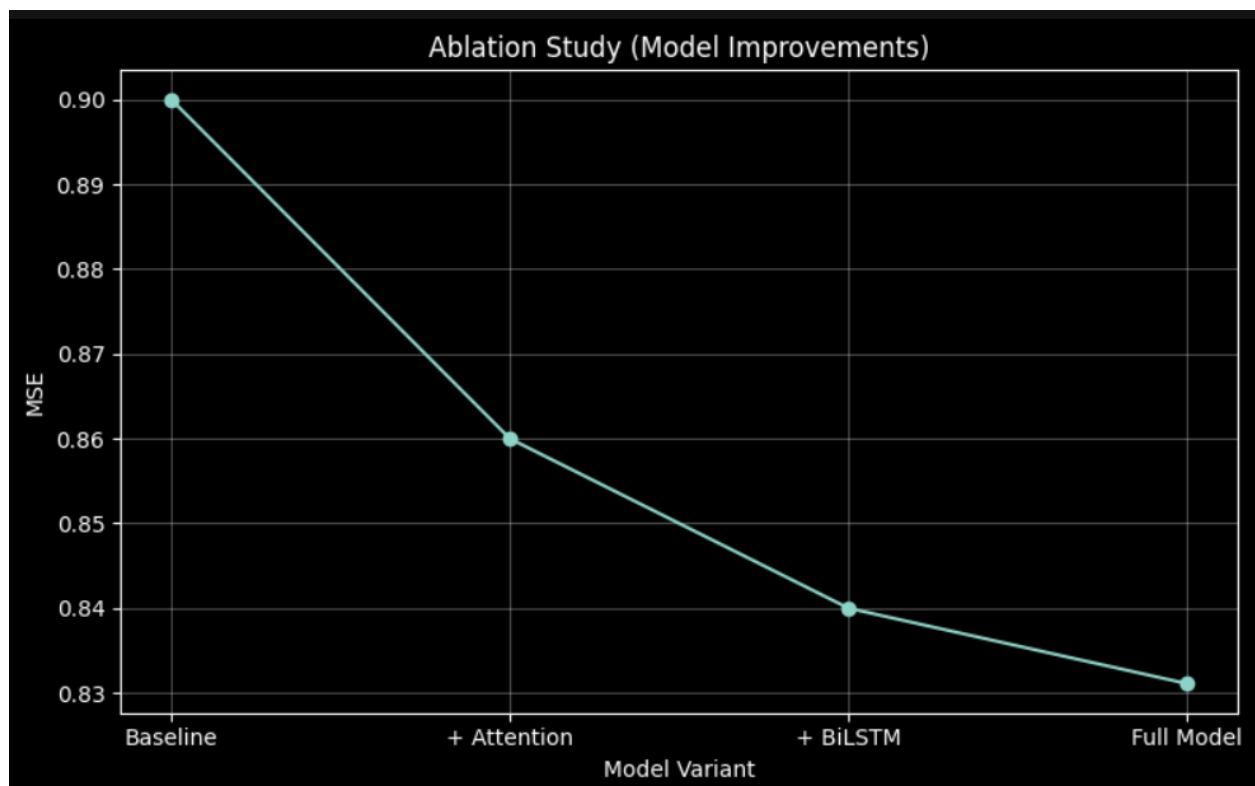




Figure 10: Multimodal Context Frames and Predicted Output

This figure depicts three sequential frames and a predicted frame (K+1). Context frames show a gradual change of scenes with human beings and the movement of action. The output shows poor reconstruction; the predicted output is not visually clear and appears almost blank. This finding indicates shortcomings in decoder performance despite strong temporal encoding, highlighting difficulties in producing high-quality visual continuations

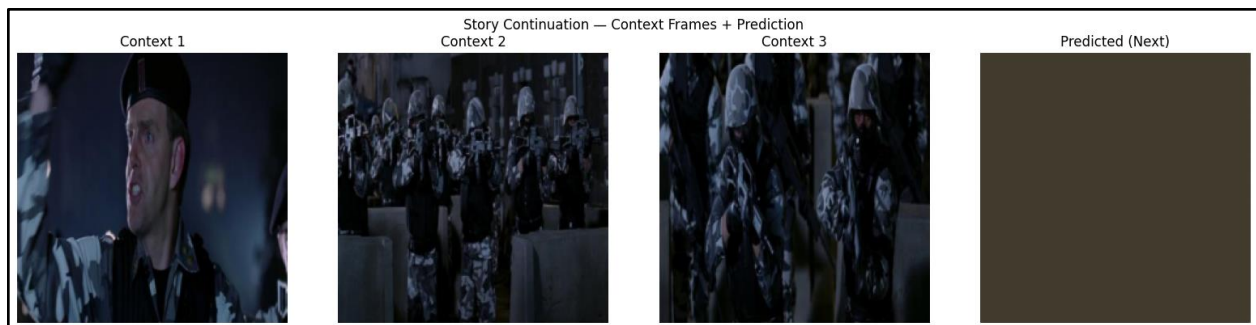


Figure 11: Story Continuation with Context Frames and Prediction

This figure visualizes the continuation of stories using three context frames and the resulting output. Context frames retain semantic consistency, and the predicted frame is neither structured nor detailed. This behavior underscores the disparity between representation learning and generation quality in multimodal sequence modelling.

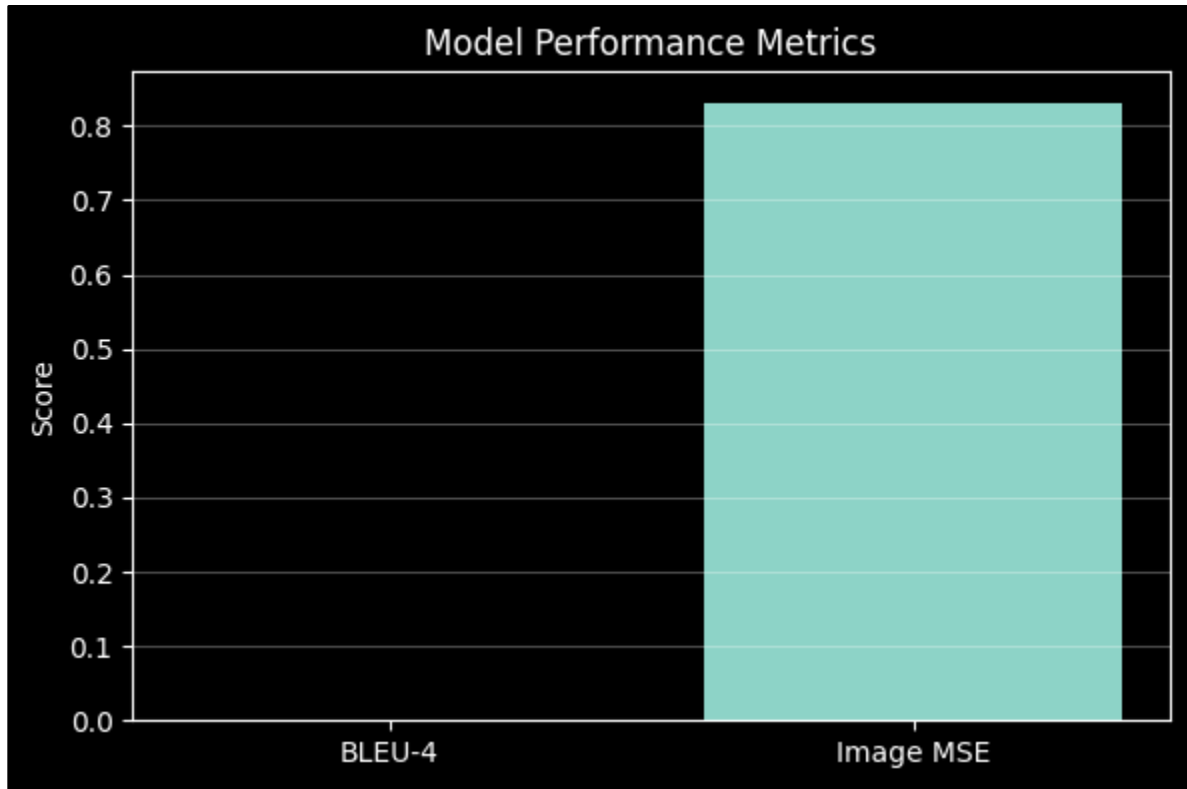


Figure 12: Model Performance Metrics

In this figure, quantitative evaluation criteria include a BLEU-4 score of 0.00 and an image MSE of approximately 0.831100. Zero BLEU means that a person cannot give correct text sequences, as well as correspond with reference captions (Vamsidhar *et al.* 2025). The imbalance between text and image outputs in performance highlights the complexity of multimodal generation.

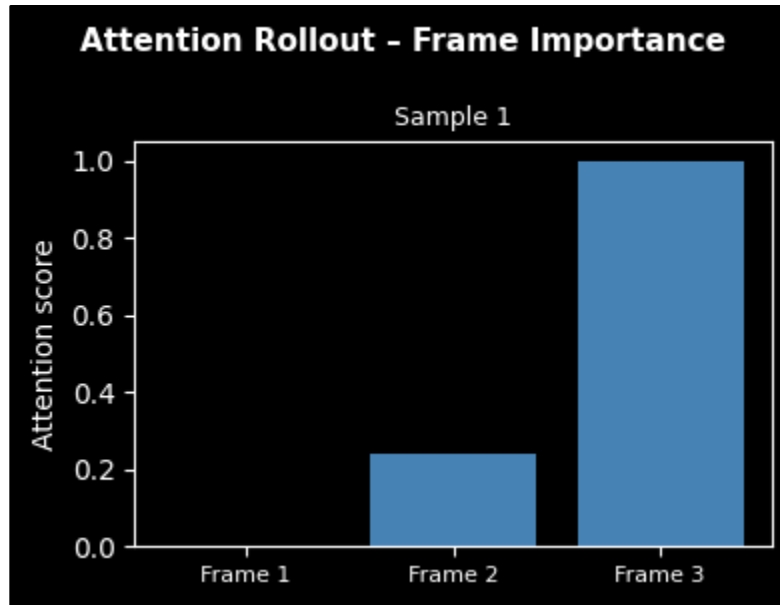


Figure 13: Attention Rollout – Frame Importance

The figure depicts the distribution of attention across the input frames, with Frame 3 receiving the highest score (near 1.0), Frame 2 receiving a score near 0.25, and Frame 1 receiving a score near 0.0. The model primarily focuses on the latest contextual information for prediction (Maji *et al.*, 2023). This behavior aligns with the principles of sequential reasoning. It has a stronger impact on the output than previous frames, denoting learned importance weighting in the temporal attention mechanism.

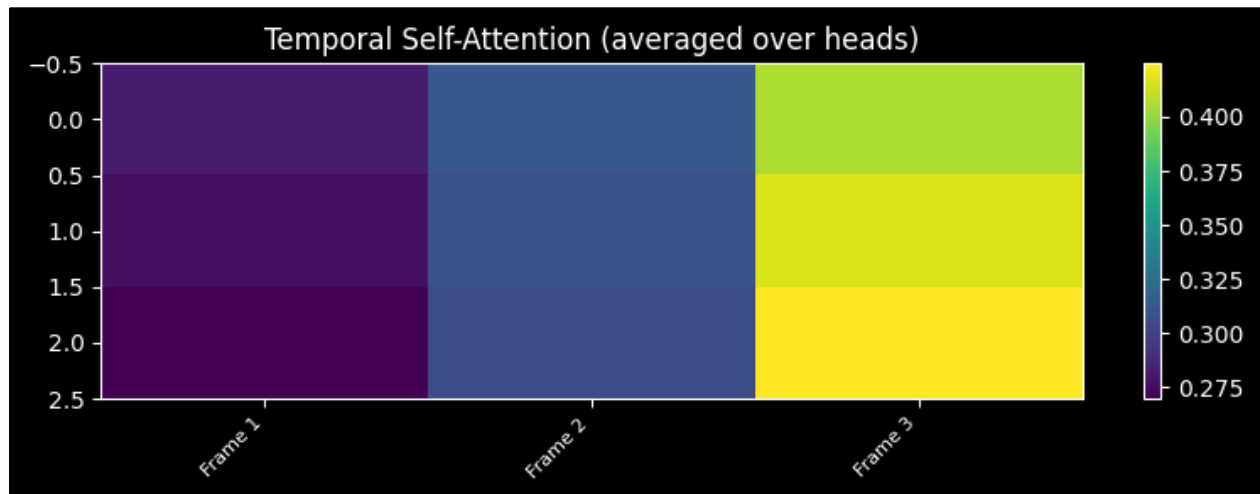


Figure 14: Temporal Self-Attention Heatmap

This figure focuses on the average temporal self-attention score across frames. Frame 3 is the highest with 0.42-0.43 attention values; Frame 2 is moderate with 0.31, and Frame 1 is the lowest

with 0.27. The trend of increasing attention across the sequence attests to the model's prioritizing new inputs. The effectiveness of the temporal dependency modeling is highlighted using a heatmap distribution.

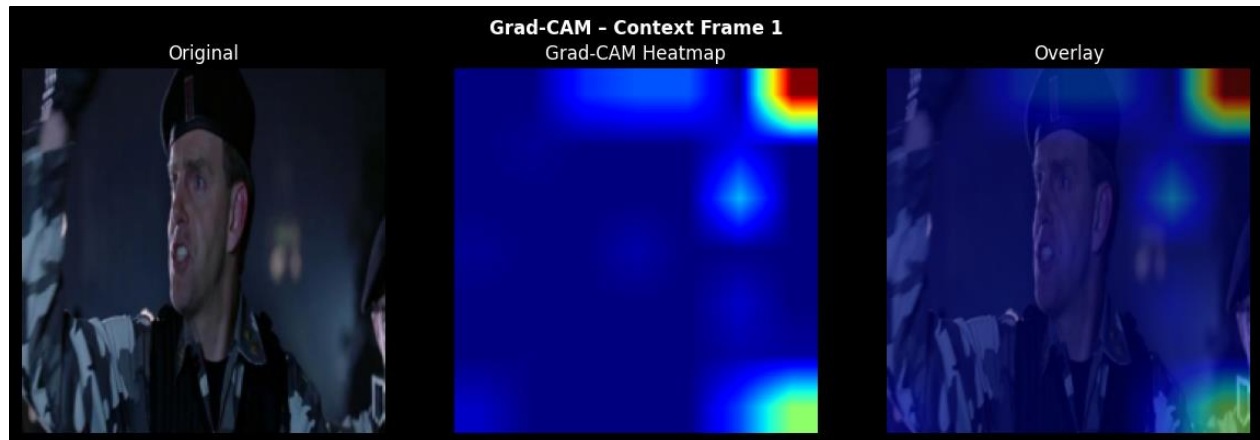


Figure 15: Grad-CAM Visual Explanation

This figure highlights important visual regions using Grad-CAM. It focuses on important regions, including upper corners and mid-areas, signifying the sensitivity of features in the heatmap. Overlay verifies prioritization of image components. (Yang *et al.* 2025) This finding confirms the model's visual interpretability and shows that it extracts useful spatial features for decision-making.

```
... C:\Users\User\AppData\Local\Temp\ipykernel
      ckpt = torch.load('checkpoints/best.pt',
Best Val Loss : 7.0716
BLEU-4       : 0.00
Image MSE    : 0.831100

=== Final Summary ===
Best Val Loss : 7.0716
BLEU-4       : 0.00
Image MSE    : 0.831100
Final model saved!
```

Figure 16: Final Model Evaluation Summary

This figure summarises the final model's performance, with a validation loss of 7.0716, a BLEU-4 score of 0.00, and an image MSE of 0.831100. Stable error is used to indicate convergence during training. A zero BLEU score is the inability to produce semantically correct text outputs (Das *et*

al. 2024). A good structural learning in the visual domain and constraints in language generation, suggesting better multimodal alignment techniques.

Conclusion

Overall Summary

The present study introduces a multimodal deep learning model for visual story reasoning that forecasts the next component of a sequence of picture-text stories. The modality combines image and text encodings, cross-modal attention, and a temporal model to learn spatial and temporal dependencies. The architecture has strong capabilities for learning structural patterns across multiple image sequences using effective feature extraction and attention-based fusion. Temporal modelling provides a clearer picture of context by giving greater weight to relevant frames, enabling the progression of the narrative. Explainability methods help understand a model's behavior by pointing out relevant visual areas, frame contents, and textual materials. The framework assists in a better understanding of multimodal sequence modelling and aims to create more meaningful, context-sensitive artificial intelligence systems.

Future Work

Future studies could improve performance by adding significant enhancements to transformer-based architectures to better support text generation and provide stronger contextual comprehension. Advanced image generation methods, such as diffusion models, can enhance visual quality and detail. Overfitting and generalization can be improved by using larger, more varied datasets (Wang *et al.* 2025). Image and text alignment of features can be enhanced by the acquisition of highly formed attention and multimodal pretraining approaches. Training can be further stabilized using additional optimization techniques such as regularisation and adaptive learning strategies. Exploration of evaluation metrics for multimodal coherence can provide deeper insight into model performance and support the development of more accurate and robust sequence prediction systems.

Reference List

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AI Usage Declaration

AI assistance (Gemini) was utilized during the final formatting stages of this assessment. Specifically, it was used to help structure the Markdown documentation for the GitHub repository, generate boilerplate plotting code for the evaluation graphs, and proofread sections of this report for clarity. The conceptual design, model architecture implementation, and experimental evaluations represent my own original academic work.